

Linear Feedback Shift Register (LFSR)

Stream ciphers require a keystream to generate a sequence of keys (bits, bytes, words...) to be combined (usually with a bit-wise XOR or modular addition) with the plaintext to produce a ciphertext.

The m -bit linear feedback shift register (LFSR) can be used to generate a binary keystream and is described by:

$$k_{m+i} = \sum_{j=0}^{m-1} c_j k_{j+i} \bmod 2$$

where $i \geq 1$, $(c_0, c_1, \dots, c_{m-1})$ are constant bits, and $(k_1, k_2, \dots, k_m)$ are the initial bits.

Example:

Find the first ten bits of the binary keystream generated by the 4-bit LFSR with constants (1,1,0,0) and initial bits (1000)

$$k_{4+1} = \sum_{j=0}^3 c_j k_{j+1} \bmod 2 = 1(1) + 1(0) + 0(0) + 0(0) \bmod 2 = 1$$

$$k_{4+2} = \sum_{j=0}^3 c_j k_{j+2} \bmod 2 = 1(0) + 1(0) + 0(0) + 0(1) \bmod 2 = 0$$

$$k_{4+3} = \sum_{j=0}^3 c_j k_{j+3} \bmod 2 = 1(0) + 1(0) + 0(1) + 0(0) \bmod 2 = 0$$

$$k_{4+4} = \sum_{j=0}^3 c_j k_{j+4} \bmod 2 = 1(0) + 1(1) + 0(0) + 0(0) \bmod 2 = 1$$

$$k_{4+5} = \sum_{j=0}^3 c_j k_{j+5} \bmod 2 = 1(1) + 1(0) + 0(0) + 0(1) \bmod 2 = 1$$

$$k_{4+6} = \sum_{j=0}^3 c_j k_{j+6} \bmod 2 = 1(0) + 1(0) + 0(1) + 0(1) \bmod 2 = 0$$

The first 10 bits of the binary keystream are: 1000100110...

Rivest Cipher 4 (RC4)

RC4 is a stream cipher that uses an 8-bit keystream to encrypt each byte of plaintext. The cipher is notable for the simplicity of the keystream generator (essentially three loops with “swaps”) which is able to produce a keystream with an estimated period of 10^{100} :

```
#initialization
for i in range(n):
    S[i]=i
    T[i]=key[i%len(key)]

#initial permutation of S
j=0
for i in range(n):
    j=(j+S[i]+T[i])%n
    S[i],S[j]=S[j],S[i] #swap
```

```
#keystream generation
i=0
j=0
while True:
    i=(i+1)%n
    j=(j+S[i])%n
    S[i],S[j]=S[j],S[i] #swap
    t=(S[i]+S[j])%n
    print(S[t])
```

Using $n=256$ and some arbitrary length “key” would generate a valid 8-bit RC4 keystream.

Example:

Encrypt the plaintext “INFO” using the RC4 keystream “0A 37 11 F2 EA...”

Converting “INFO” to hex by ascii: 49 4E 46 4F

Bit-wise XOR each byte with the keystream elements:

```
49 4E 46 4F
0A 37 11 F2
43 79 57 BD
```

The ciphertext “437957BD” can be decrypted by bit-wise XOR of each byte with the same sequence of keystream elements.

Exercises:

1. The first ten bits of the binary keystream generated by the 4-bit LFSR with constants (1,1,0,0) and initial bits (1000) are: 1000100110...
Find the next 10 bits in the keystream and show that this keystream has period 2^4-1 .
2. Encrypt the hex string "DEAD" using the RC4 keystream "12 A5 39..."
3. Calculate the *initial permutation of S*, and the first three elements in the keystream of a "mini" RC4 with $n=4$ and $\text{key}=[2,1]$.

Answers:

1. 10001001101011110001...
2. CC08
3. $S = \{1, 2, 3, 0\}$
"11 00 01..."